DigitalTwin.ai

Detailed Business Proposal · Accenture Innovation Challenge 2026, Problem Track 4 · Team HipHipHooray · Sagar Sahu, Priyansh Goyal · IIT Kharagpur

The one-line claim. A vehicle assembly line already emits enough data to say *which station is costing you cars right now*, *which one is about to*, and *what to do about it* — on a line where nearly a fifth of the stations have no sensors at all. We built that loop, ran it for **903 shifts**, and every number in this document is measured against held-out ground truth with its source named.

15.5%	0.025	70.5%	903
of forming-bottleneck warnings name a station with **no sensors at all** [15.0–15.9], n=29,060	calibration error (ECE) after fitting, from 0.479 600 held-out samples	of alarmed tools get an **actionable** diagnosis [67.9–72.9], n=1,246	shifts replayed through the live loop 86,742 decisions recorded

1 · Problem framing

A plant manager can already tell you last month's OEE. What no system reliably tells them is the question that actually decides output: **at 14:20 today, which single station is holding the line back, and is it worth walking to?**

Three things make that hard, and they are the three the brief names:

- **The constraint moves.** In our data it changes station roughly **20 times per shift**. A weekly report averages that away, and an average describes no moment of the shift it covers.
- **Utilisation lies.** A station that is blocked by its neighbour and a station that is genuinely slow both read ~95% busy. Ranking by busyness sends people to the victim.
- **The data is uneven.** Real lines mix legacy and modern equipment. Some stations are richly instrumented; others emit nothing at all and rely on a human with a clipboard.

The gap is not sensing and it is not dashboards. It is the step from *describing* the line to *deciding* on it — and doing so honestly enough that a supervisor keeps trusting the system after the third false alarm.

2 · Solution design

2.1 A twin, not a shadow

The distinction that matters (Kritzinger's taxonomy): a *model* is offline, a *shadow* receives data one-way, a *twin* closes the loop. Our prototype ingests, detects, ranks, prescribes and re-reads on a timer, and the causality is **enforced structurally**: the detector is handed a view of the run truncated at `now`, so it cannot read the future even by accident. Verified identical verdicts at 13/13 timepoints against the full-history run.

2.2 Four tiers of knowledge, never mixed

TIER	WHAT IT IS	WHERE IT COMES FROM
A — measured	The station reports it	PLC state, scans, tool telemetry
B — inferred	Bracketed between neighbours	Boundary timestamps, buffer slope
C — attested	A person says so	Manual checklist entry, with its entry latency
D — predicted	Projected under current flow	Buffer countdown

Every value on screen carries its tier. An operator must never see an inferred number and a measured one in the same font.

2.3 How the constraint is found

Rank by **effective cycle time** — work time divided by availability — not by busyness. Work time excludes seconds the station spent blocked or starved, so a victim is not charged for its neighbour's fault. Each station is compared to **its own early baseline**, never to takt: a station drifting 54→57 s is abnormal even while comfortably inside a 60 s takt.

2.4 From detection to prescription

The twin names the action and what ignoring it costs. The action vocabulary is a small library keyed on fault classes the detector itself separates; the **ranking and the cost are computed**, never retrieved. That is the step from descriptive → diagnostic → predictive → **prescriptive**.

> **The alert contract.** An alert may be raised only if it carries all five of: the ranked candidate and its margin; the evidence that produced it; a persistence estimate; the recommended action; and the expected cost of not acting. **An alert that cannot state its evidence is suppressed, not downgraded** — a quiet alert with no evidence is still noise, and noise is what erodes floor trust. In the recorded run, 7 alerts were suppressed by this rule.

2.5 What the twin needs from a plant, and what it must never touch

The integration surface is deliberately narrow: **five streams a plant already produces.** Every module downstream is written against this contract rather than against our simulator, which is why the same code runs on our data and would run on a plant's.

WHAT THE LIVE TWIN READS	WHERE A PLANT ALREADY HAS IT
Unit scans — VIN, station, timestamp	Barcode/RFID readers at station boundaries; already mandatory for traceability
Station state transitions	PLC state tags, already historised for OEE reporting
Buffer levels	Conveyor counters and occupancy sensors
Tool readings — torque, angle, current	Nutrunner controllers over Open Protocol / OPC-UA
Manual checks — result and entry latency	The paper or tablet checklist at an uninstrumented station

Our simulator also emits andon, rework and calendar logs, and those ship with the sample data, but the live loop does not consume them today — so they are not counted above.

Nothing on that list requires new hardware. The twin's day-one deployment buys no sensors; the retrofit schedule in §5 is what it earns later, once the twin has measured which dark station is actually costing money.

> **The boundary is enforced, not promised.** Exactly two modules in the codebase may write anything, and both write only to the twin's own store. **No module may import a network client** — not `socket`, not `opcua`, not `pymodbus` — so the twin cannot reach a PLC even by mistake. This is checked by a test that walks the syntax tree of every module on every run, and we verified the test fails by planting a PLC write and confirming it was caught. The read-only claim is therefore a property of the build, not a promise in a document.

> **What stands between this and a live line: one adapter.** We have not built a historian or MES connector — the prototype reads these streams from files. That is the honest gap, and it is deliberately the *only* one: because every module reads the contract and not the simulator, connecting a real plant means writing one module that subscribes to that plant's historian and emits these tables. Nothing downstream changes. **Shadow mode already works today** — point the replay driver at a plant's exported logs and the entire twin runs, with no live connection of any kind. That is Phase 0 of §7, and it is the phase we can already execute.

3 · Target users

Three users, three resolutions, **one model** — and we prove they are one rather than asserting it.

USER	NEEDS	WHAT WE BUILT
Floor supervisor	Real time, in the moment	ISA-101 display: grey base, colour only on deviation. A well-designed plant HMI is visually quiet during normal operation. Dark stations are drawn hatched, not hidden.
Plant manager	Weekly planning	A **constraint-occupancy distribution**, deliberately *not* an average — the constraint moves ~20×/shift, so a mean describes no moment of it. "S20 held the line 21.3% of week 1" is a scheduling and capex input; a mean is not.
Leadership	Investment case	Value, coverage and the evidence table — every figure naming the file that produced it. Colour is allowed here: ISA-101 governs a plant HMI, not a business dashboard.

The reconciliation test. Each level totals independently and is compared: **6,730 constraint-minutes and 4,431 vehicles, identical across supervisor records, manager weeks and leadership.** That is how we show three views are one twin instead of three dashboards.

4 · Business case and impact

4.1 Detection quality, measured against economic ground truth

Ground truth is not "which station looked busiest". It is the station whose speed-up **actually produces more cars**, measured by re-running the line under common random numbers. That makes the benchmark economic rather than definitional.

METHOD	TOP-1	TOP-2	REGRET (CARS/BLOCK)
Effective cycle time (ours)	43.1%	59.4%	**1.309**
Active period (Roser 2001)	46.0%	57.9%	1.348
Utilisation (the common baseline)	35.1%	48.0%	1.477

Top-1/top-2 on 202 strong-constraint blocks; regret on all 958 blocks. Source: `results/eval_v5.csv`.

What we may and may not claim. Paired McNemar tests on the same blocks: we **significantly beat the utilisation baseline ($p = 0.0025$)**, and we are **statistically tied with the active-period method ($p = 0.45$)**. We therefore claim the first and never the second. Comparing independent confidence intervals would have been the wrong test for same-block data and would have let us overclaim.

4.2 Why regret, and not accuracy

Two independent findings point the same way. First, the sensitivity labels carry a **0.79-car noise floor**, while the median margin between the best and second-best station is only 0.50 cars — so the top-1 label survives noise in only ~50% of blocks and is close to a coin flip. Second, across four line topologies **top-1 collapses from 44.5% to 10.6% while regret stays between 1.208 and 1.431**. The operationally meaningful metric transfers; the argmax metric does not.

4.3 Value, stated honestly

VALUE LINE	STATUS
Fewer cars lost to a mis-identified constraint (regret 1.309 vs 1.477 utilisation, on a 2.271-car ceiling)	**measured**
Earlier warning: buffer countdown, 178 warnings, median error +0.57 min	**measured**

Avoided false scrap: transducer-drift tools flagged for recalibration, not repair	**measured**
Sensor spend avoided: 15.5% of forming warnings need no sensor at that station	**measured**
Lead-time reduction under CONWIP release control	**NOT measured — excluded**

One value line has been deliberately removed. An earlier draft led the business case with "same throughput, 36% lower lead time, zero capital expenditure". No file in our results produces that number, so under our own rule — *a number is measured with a named source, cited with a named reference, or it does not appear* — it does not appear. We would rather present four defensible lines than five with one that dies under a single question.

5 · Handling the real-world complexities

#	COMPLEXITY	HOW WE ANSWER IT	STATE
1	Uneven sensor coverage; manual checklists	Dark stations are bracketed between neighbours and named directly by buffer slope — **15.5% of all forming warnings name a station with no sensors** [15.0–15.9], n=29,060. Manual checks enter as Tier C *attested* data carrying entry latency. **And we tested whether the checklist is honest**: across 31,329 entries it passes 96.51% of vehicles, yet **835 of those it passed went on to fail end-of-line — a 2.76% escape rate**, with a mean 4.8-minute lag before a human records anything at all. A checklist reading near-100% against a non-zero EOL failure rate is measuring compliance with the checklist, not quality.	Demonstrated
2	Multi-causal, intermittent root causes	Separated by **scope** — who else is affected and who is not — rather than by statistics. The prototype separates fault classes (slowing, breakdown, starved, blocked). **Operator variation is deliberately excluded on ethical grounds:** we measure the station, never the person. That is a design choice, stated, not an oversight.	Partly demonstrated; scope engine designed
3	PLC risk; retrofits only in maintenance windows	**The twin advises and never writes to line control** — a read-only boundary in ISA-95 terms, shown on every screen. Sensing additions mount externally and publish on a separate network, so no PLC program changes and therefore no re-validation. The retrofit list itself is **a window-dated schedule, not a wishlist** — ranked by exposure closed per rupee, split across two shutdowns, with the cost of deferring stated explicitly (~₹101,560 carried for six months).	Demonstrated
4	Early defect surfaces late	Onset is read **backwards off the CUSUM accumulator**, then the VIN thread lists exactly which vehicles carry it, **partitioned by where they are** — a car on the line is a rework instruction, one that has shipped is a customer event. Onset dated to **+2 minutes** of truth.	Demonstrated
5	Three stakeholder views	One record stream, three resolutions, with the reconciliation test passing exactly.	Demonstrated
6	Scaling: layout, vintage, sensor maturity	Measured across four topologies including a parallel-server pair that breaks the pure-series assumption. See §6.	Demonstrated (layout, sensor maturity); vintage unmodelled
7	False alarms erode trust	Confidence is **calibrated, not asserted** (ECE 0.479→0.025), and every alert is logged with a human confirm/override so precision can be shown over time. **48.6 alerts/shift** against the ISA-18.2 budget of <150.	Demonstrated

> **Against the brief's own reference parameters.** The PS assumes "a majority of stations well-instrumented, a meaningful minority reliant on manual checks" across 30–50 stations spanning body, paint and final assembly. On our 40-station segmented layout, body + final assembly — the 30 stations where per-unit measurement is meaningful — sit at **75.3% instrumented, 24.7% on manual checks: an exact match.** Paint sits at 90% dark *by design*, since booth-level sampling is how paint shops work everywhere — a property of the process, not a coverage gap we failed to close.

6 · Does it survive a different line?

LINE	N BLOCKS	TOP-1	REGRET	CEILING
L1 — 20 stations, 1 merge *(reference)*	191	44.5%	1.208	4.272
L2 — 30 stations, 2 merges	94	10.6%	1.218	1.705
L3 — 20 stations + parallel pair	94	13.8%	1.431	1.870
L4 — 15 stations, 1 merge	96	21.9%	1.221	2.096

Source: `results/transfer_eval.csv`. L2–L4 are 12 runs each, so intervals are wide (L2 top-1 [5.9, 18.5]); these are directional, not definitive.

Sensor maturity. Running the entire pipeline from boundary scans with **zero PLC state tags** costs **+1.683 cars/block on L1** but only **+0.021 on L2**. State tags matter enormously where a strong constraint exists and almost not at all on a balanced line — so **instrument the lines with the most to gain first**. That is a more useful retrofit input than a single averaged degradation.

7 · Phased roadmap

PHASE	WHAT HAPPENS	GATE TO PASS BEFORE THE NEXT PHASE
0 · Shadow weeks 1–4	Subscribe to the streams the plant already produces and discards. No writes, no PLC changes.	Twin reproduces the plant's own throughput numbers
1 · One supervisor weeks 5–10	One line, one shift, one screen. Every alert confirmed or overridden by a person.	Running precision stable over ≥20 shifts; alerts within the ISA-18.2 budget
2 · Floor months 3–6	All shifts on that line. Manager view opens for planning.	Reconciliation holds; supervisors use it unprompted
3 · Retrofit next window	Costed sensor list installed at a scheduled shutdown, ranked by exposure closed per rupee.	Measured coverage improvement matches the prediction
4 · Second line months 6–12	Transfer. Report the commissioning curve honestly, not a yes/no.	Regret within the range measured across our four topologies
—	**Never: closed-loop write to line control.**	—

8 · Key risks and mitigations

RISK	WHY IT IS REAL	MITIGATION
False alarms erode trust	The brief names it; it is how these systems die	Calibrated confidence (ECE 0.025), the five-field contract with suppression, and a ledger showing running precision. 48.6 alerts/shift vs a 150 budget
Overconfidence in a weak signal	We have already been caught by this once	We measured our own overtake-risk mechanism at **5.9% correct [1.0–27.0], n=17** against 70–100% stated confidence, declared it failed, and removed it from the live path. The buffer countdown replaced it

Top-1 accuracy is a misleading target	0.79-car label-noise floor; argmax survives jitter in ~50% of blocks	Regret is the headline metric; top-1 reported with Wilson intervals alongside
Transfer may not hold	L2–L4 are 12 runs each	Stated as directional. Regret held 1.21–1.43 across four topologies; the commissioning curve is reported, not asserted
Touching production	Regulated, safety-certified control	Read-only by construction and **enforced by a test that walks every module's syntax tree**: two modules may write, and only to our own store; no module may import a network client. Verified by planting a PLC write and confirming the test caught it
Modelling people	Operator variation is a named cause and a real hazard	Excluded by choice. We measure the station, never the person, and say so

9 · What we deliberately did not do

- **Equipment vintage** is one of the three scaling axes the brief names and we have no data axis for it. Unmodelled, and stated.
- **Operator variation** — excluded on ethical grounds, as above.
- **The CONWIP lead-time figure** — removed for want of a source file.
- **Closed-loop control** — outside what any plant grants a prototype.
- **A historian/MES adapter** — the prototype reads its input streams from files, not from a live plant connection. Deliberately the only gap between this and Phase 0 on a real line, and narrow by design: see §2.5.

Every one of these is a decision we can defend, which is worth more than a claim we cannot.

All figures traceable to `results/` and `results/twin.db` (903 shifts, 86,742 recorded decisions). Reproduce with `scripts/query_twin.py`, `scripts/eval_v5.py`, `scripts/eval_transfer.py`, `scripts/fit_calibration.py`. Repository: github.com/sagar2907/HipHipHooray_IIT_Kharagpur_DigitalTwin.ai_Prototype